

Mathematics - NCERT

Time Allowed : 60 Minutes _____
Maximum Marks : 120

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. Paper contains 12 questions. Each question in this section contains 10 marks.

- Q1:** How many 2 digit even numbers can be formed from the digits 1, 2, 3, 4, 5 if the digits can be repeated?
- Q2:** Given 5 flags of different colours, how many different signals can be generated if each signal requires the use of 2 flags, one below the other?
- Q3:** If $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$, find x .
- Q4:** Find the value of n such that
- $$\frac{{}^nP_4}{{}^{n-1}P_4} = \frac{5}{3}, n > 4$$
- Q5:** From a committee of 8 persons, in how many ways can we choose a chairman and a vice chairman assuming one person can not hold more than one position?
- Q6:** In how many of the distinct permutations of the letters in MISSISSIPPI do the four I's not come together?
- Q7:** Determine the number of 5 card combinations out of a deck of 52 cards if there is exactly one ace in each combination.
- Q8:** A bag contains 5 black and 6 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected.
- Q9:** Find the number of words with or without meaning which can be made using all the letters of the word AGAIN. If these words are written as in a dictionary, what will be the 50th word?
- Q10:** In how many ways can 5 girls and 3 boys be seated in a row so that no two boys are together?
- Q11:** The English alphabet has 5 vowels and 21 consonants. How many words with two different vowels and 2 different consonants can be formed from the alphabet ?
- Q12:** From a class of 25 students, 10 are to be chosen for an excursion party. There are 3 students who decide that either all of them will join or none of them will join. In how many ways can the excursion party be chosen ?